

intelligence has led to a dramatic transformation in computing, with an insatiable hunger for high-performance computing power, at the cost of power consumption and infrastructure requirements.

Traditional AI processing requires high-end GPUs that demand energy, unsuitable for edge applications. SandLogic started deploying AI models on compact devices, reducing reliance on cloud computing or energy-intensive processors.

Founded in 2018 by Kamalakar Devaki, Jesudas Fernandes, Radhika Kanigiri, and Ravi Kumar Rayana, Bengaluru-based SandLogic developed ExSLerate, a low-power AI co-processor IP designed to bring efficient AI computation to consumer devices, medical devices, drones, IoT, and other edge applications.

Unlike traditional processors, which act as external accelerators requiring continuous interaction with the host processor, ExSLerate boasts of minimising memory transactions by 90%. “This results in significantly lower power consumption while boosting efficiency. We have filed a patent for this innovation,” says Kamal.

“This is particularly useful in edge computing applications, where power efficiency and real-time AI decision-making are critical. Our proprietary software stack, EdgeMatrix, which sits on existing processors and upcoming chips packing ExSLerate IP, improves the co-processor’s performance by optimising token generation for AI models,” he elaborates.

ExSLerate can be clustered for larger AI workloads without requiring extensive redesign. The co-processor is compatible with multiple neural network architectures, including convolutional neural networks (CNNs) and transformers. The startup has integrated its edge AI platform, EdgeMatrix.io, into the design to optimise AI workflows on the chip itself. It packs all the required software tools



(L to R) Ravi Kumar Rayana (AI and Enterprise), Jesudas Fernandes (AI Architect), Kamalakar Devaki (Founder), and Radhika Kanigiri (Co-Founder)

to push AI workloads to the chip, including compilers, parsers, and various run-time optimisers.

“Our architecture supports dynamic sparsity, which improves performance and throughput. It is designed to accommodate all types of neural networks, including CNNs and Transformers, and supports quantisation down to INT4 for additional performance gains. For context, comparable chips in this space—such as Google’s Edge TPU and Xilinx B10-1024—typically consume between 1 and 5 watts of power. In contrast, our chip delivers 15 trillion operations per second (TOPS) at just 2 watts while offering all the aforementioned advancements,” explains Kamal.

To validate its design, SandLogic has partnered with the Centre for Development of Advanced Computing (CDAC) through the Indian government’s Chips to Startup (C2S) programme. It also participated in the Qualcomm mentorship programme and is part of SFAL. SandLogic recently collaborated with Sasken Technologies for various chip-related initiatives within India and beyond.

The startup is choosing to forego the full-scale semiconductor fabrication model and is instead taking an IP-driven approach. It intends to license its AI co-processor design to semiconductor manufacturers rather than fabricate chips in-house.

Kamal cites the capital-intensive nature of semiconductor fabrication

as the reason for this strategy. “By focusing on IP licensing, we can scale faster without the burden of fabrication. Additionally, most of our potential customers prefer to integrate the design into their own fabrication processes rather than purchase pre-made chips.”

For validation and proof-of-concept (PoC), SandLogic will manufacture a limited number of KRSNA test chips, primarily for PoC demonstrations. These chips will be tested by industry partners to showcase real-world performance before commercialisation. Kamal revealed that the startup has caught the eye of multiple industry players, including the Defence Research and Development Organisation (DRDO).

While Kamal projects that commercial sales will begin in June 2026, SandLogic’s biggest challenge is securing funding and strategic partnerships for validation and market expansion. The company is currently in the middle of a Series A funding round to raise \$30 million. The funding will be spent on technology development, validation, establishing distribution channels, and expanding industry outreach efforts. The startup is also looking to partner with semiconductor fabs, embedded system integrators, and device manufacturers.

While SandLogic’s immediate focus is on small-form-factor AI processing, in the long run, Kamal and his team are looking to support next-generation AI models such as liquid field models and recurrent kernel-weight vectorisation (RKWV). The startup is also considering open-sourcing the base version of ExSLerate once the technology is validated, to foster wider industry adoption. **EFY**

—Yashasvini Razdan